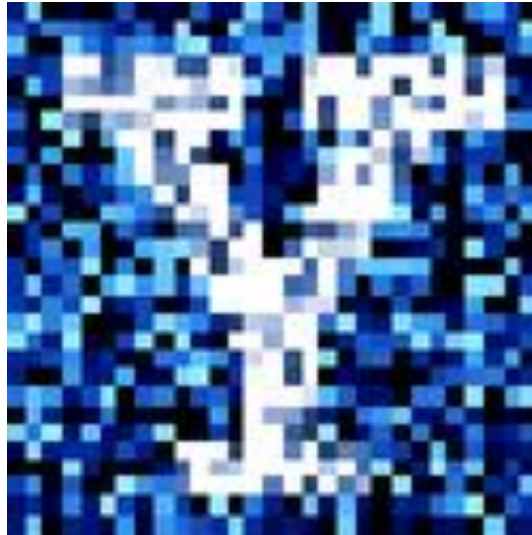


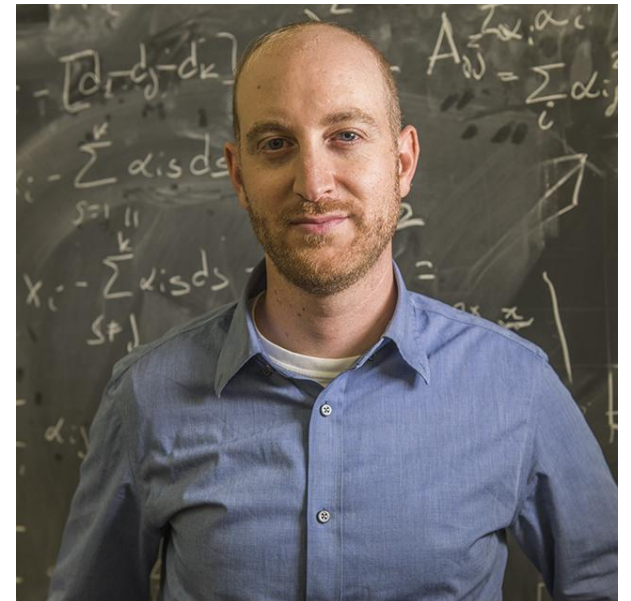
# YData: Introduction to Data Science



Class 01: Introduction

# Credit

- YData was originally based on UC Berkley's Data 8:  
<http://www.data8.org/>
- A lot of the material we will use was written by Ethan Meyers (thank you, Ethan!).



# Overview

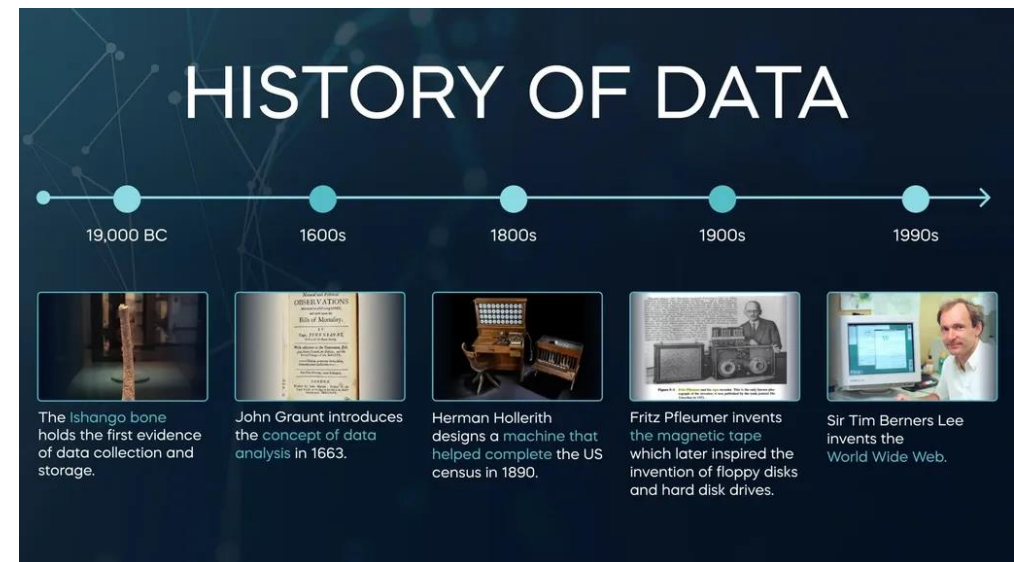
## Course overview

- Introductions
- Syllabus and logistics



## What is data science?

- Brief history of data science



# Office hours and contact information

Yuejie Chi (she/her)

Email: [yuejie.chi@yale.edu](mailto:yuejie.chi@yale.edu)

Office hours:

- Wednesdays, 1:30-2:30pm
- By appointment

Office: Kline Tower, room 1249

# Teaching Assistants

## Preceptor

- Shivam Sharma: [shivam.sharma@yale.edu](mailto:shivam.sharma@yale.edu)

## Teaching Fellows

- Yidan Mai: [yidan.mei@yale.edu](mailto:yidan.mei@yale.edu)
- Minoli Daigavane: [minoli.daigavane@yale.edu](mailto:minoli.daigavane@yale.edu)
- Siyu Chen: [siyu.chen.sc3226@yale.edu](mailto:siyu.chen.sc3226@yale.edu)

## Undergraduate Learning Assistants

- Kyle Levesque: [kyle.levesque@yale.edu](mailto:kyle.levesque@yale.edu)
- Bruno Kai Ong: [brunokai.ong@yale.edu](mailto:brunokai.ong@yale.edu)
- Sloane Huey: [sloane.huey@yale.edu](mailto:sloane.huey@yale.edu)
- Christian Baca: [christian.baca@yale.edu](mailto:christian.baca@yale.edu)



TA office hours will be posted on the calendar on Canvas

# Introductions



Let's do some quick introductions

Create groups of 3-4 people:

- Your name and preferred gender pronouns
- Your major/grad dept (research area)
- Why you are interested in this class
- Anything else you would like to share with your group

About this class

# Topics covered

What is Data Science?

Python basics

Descriptive statistics

Array computations

Manipulating data tables

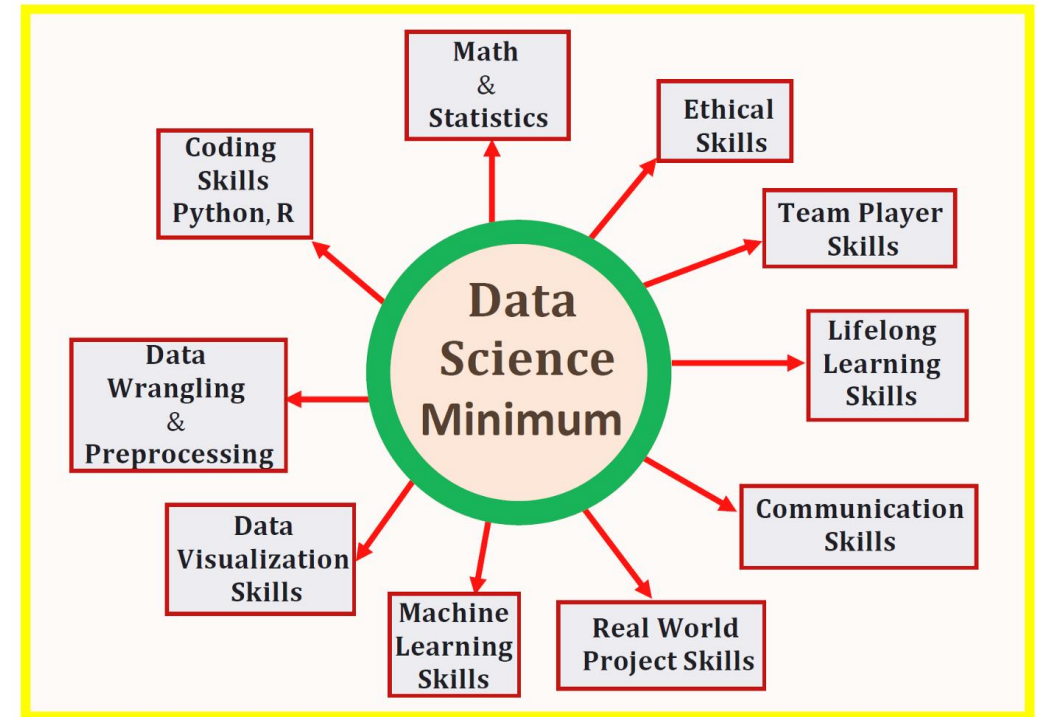
Data visualization

Mapping

Text manipulation and data cleaning

Statistical perspective: hypothesis tests and confidence intervals

Machine learning perspective: supervised and unsupervised learning





# Tentative plan for the semester: **subject to change!**

| Week       | Date           | Topic                                                  | HW Assigned            | HW Due |
|------------|----------------|--------------------------------------------------------|------------------------|--------|
| 1          | Jan 13         | Class overview                                         | 0                      |        |
| 1-2        | Jan 15, Jan 20 | Introduction to Python                                 | 1                      | Jan 25 |
| 2-3        | Jan 22, Jan 27 | Descriptive statistics and plots                       | 2                      | Feb 1  |
| 3-4        | Jan 29, Feb 3  | Array computations                                     | 3                      | Feb 8  |
| 4-5        | Feb 5, Feb 10  | Tables and data manipulation                           | 4                      | Feb 15 |
| 5-6        | Feb 12, Feb 17 | Data visualization                                     | 5                      | Feb 22 |
| 6-7        | Feb 19, Feb 24 | Review and midterm exam                                |                        |        |
| 7          | Feb 26         | Mapping                                                | 6                      | Mar 8  |
| 8          | Mar 3-5        | For loops and writing functions                        | 7                      | Mar 29 |
| 9-10       |                | Spring break                                           |                        |        |
| 11         | Mar 24-26      | Intro to statistical inference and hypothesis tests    | 8                      | Apr 5  |
| 12         | Mar 31, Apr 2  | Hypothesis tests continued and confidence intervals    | Draft of final project |        |
| 13         | Apr 7-9        | Machine learning: classification                       |                        |        |
| 14         | Apr 14-16      | Machine learning: regression and unsupervised learning | 9                      | Apr 19 |
| 15         | Apr 21-23      | Ethics and Review                                      | Final project          |        |
| Final exam | Apr 30 2pm     | In person final exam                                   |                        |        |

# Learning goals

## 1. Understand concepts in data science

- Learn basic computational skills for analyzing data
- Understand concepts in statistics and machine learning

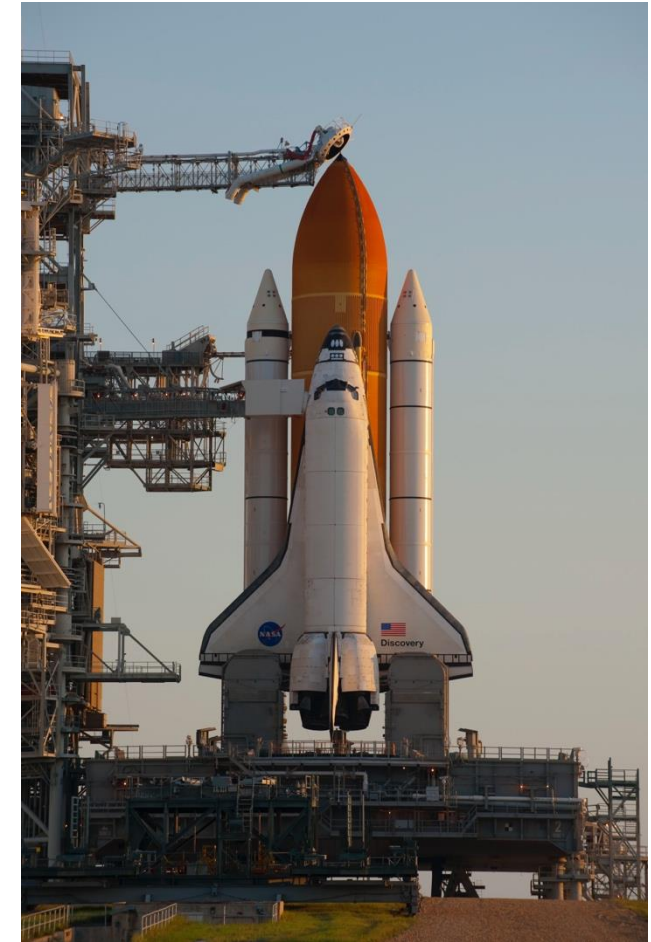
## 2. Gain practical data science skills applicable to any domain

## 3. See how data science analyses can be applied to real-world data from a variety of domains

- There will be ~weekly readings on data science related topics

There are no prerequisites for this class

- E.g., no prior knowledge of statistics or programming is required



# Course structure

Two lectures per week

Weekly Python practice sessions (optional)

Weekly homework assignments

A class project

Weekly drop-in office hours to get help on homework (see Canvas)

Midterm and final exam

I recommend coming to class in-person.  
As a backup, the class should usually be recorded and  
available on-line after the lecture.  
I will post the slides and in-class code.

# Python practice sessions

Shivam is hosting one-hour practice sessions each week

- Each session will be offered at several different times each week
- Please fill in class survey to let us know what times work for you

Sessions will be a great opportunity to practice Python and get your questions answered!

Highly recommended to attend these sessions

- Attendance is optional
- If you score below a median cutoff score on the exams, a point will be added to your score for each weekly practice session you attend (up to 5 points)



# Class readings

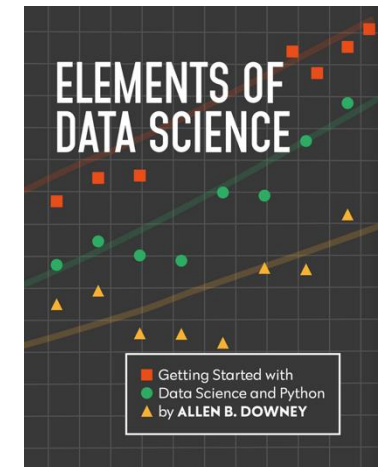
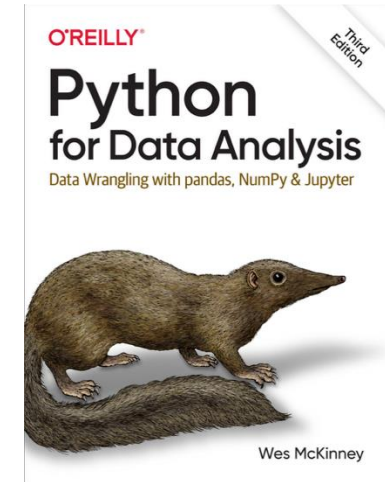
There will be short readings on data science topics approximately every week

- Links to the readings will be on the homework

Readings will also be taken from:

- Brett M (2020). [Data Science for Everyone: course text](#)
- McKinney (2022). [Python for Data Analysis, 3E](#)
- Downey (2024). [Elements of Data Science](#)
- Other sources posted to Canvas/on the Internet

Resources related to programming will also be posted on Canvas under the appropriate class



# Assignments and grades

## 1. Homework problem sets (48%)

- Exploring concepts and analyzing data using Python
- Weekly: 9 in total

## Homework policies

- You may discuss questions with other but the work you turn in must be your own!
- Homework released by class on Thursdays and are due at 11pm on Sundays
  - (with a 59 minute grace period)
- Late homework (90%) credit if turned in by 11pm on Monday
  - For any other extensions a Deans Excuse is needed
- Lowest scoring homework will be dropped

# A typical homework assignment

## 3. Basic statistics: Differences between Universities

**Question 3.1 (8 points)** Suppose you'd like to *quantify* how *dissimilar* two universities are, using three quantitative characteristics. The US Department of Education data on [Yale](#) and [Cal](#) describes the following three traits (among many others):

| Trait                                | Yale   | Cal    |
|--------------------------------------|--------|--------|
| Average annual cost to attend (\$)   | 16,341 | 15,240 |
| Graduation rate (percentage)         | 97     | 94     |
| Socioeconomic Diversity (percentage) | 20     | 23     |

You decide to define the dissimilarity between two universities as the maximum of the absolute values of the 3 differences in their respective trait values.

Using this method, compute the dissimilarity between Yale and CAL. Name the result `dissimilarity`. Use a single expression (a single line of code) to compute the answer. Let Python perform all the arithmetic (like subtracting 97 from 93) rather than simplifying the expression yourself. Be sure that the final answer is printed out in the .pdf file you upload to show that you have the correct answer.

Hint: The built-in `abs()` function takes absolute values.

```
dissimilarity = ...
```

# Assignments and grades

## 2. Project (10%)

- A draft of your class project is due 2/3<sup>rd</sup>s of the way through the semester
- You will give and receive feedback from your peers
- Final version of the project will be turned in at the end

The only thing worse than an exam is a coding exam on paper...  
But we haven't found a good alternative for this course.  
It's also a skill for job interviews.

## 3. Exams (40% total)

- Midterm: 2/27 during the regular class time (15%) [subject to change]
- Final Exam: Saturday 5/3 at 9am (25%) [Date might not be final]
- To take the class you must be able to attend these exams at their scheduled times!

## 4. Participation (2%)

- Active asking and answering questions on Ed Discussions



# Assignments and grades

## 2. Project (10%)

- A draft of your class project is due 2/3<sup>rd</sup>s of the way through the semester
- You will give and receive feedback from your peers
- Final version of the project will be turned in at the end of the semester

## 3. Exams (40% total)

- Midterm: October 9<sup>th</sup> during the regular class time (15%)
- Final Exam: Monday December 17<sup>th</sup> at 2pm (25%)
- **To take the class you must be able to attend these exams at their scheduled times!**

## 4. Participation (2%)

- Active asking and answering questions on Ed Discussions

# Grade distribution

Grade cut-off are

- A [94-100], A- [90-94), B+ [87-90), B [80-84), etc.
  - I might slightly modify these downward if the class too hard

No strict grade distribution but roughly:

- 25% A, 25% A-, 25% B+, 25% everything else

Students generally score high on the homework ( $> 90$ ) and exam scores tend to be lower ( $\sim 80$ )

If an exam is too hard, I sometimes curve them by adding "free points"

- E.g., if an exam is out of 85 points, I might add a free 15 bonus points so the exam is out of 100

**Please focus on learning rather than on grades!**

# Accommodations and Academic honesty

**Accommodation:** please let me know if you have accommodations for homework and/or exams

Plagiarism/cheating

- [Yale's Academic Integrity Statement](#)

You are allowed to talk with others about the homework, but the work you turn in must be your own

- Do not share answers
- Do not copy answers off the Internet
- Do not look at past year's homework

# ChatGPT and other LLMs

You can use LLMs as a reference

- E.g., "What is the function to do x?"
  - i.e., ok to use it like Google/Stack Overflow

## **Do not use it to answer homework questions**

- i.e., do not type a homework question into ChatGPT
- Do not have it complete code you started writing

If it appears your homework answers were generated by ChatGPT or another LLM (Claude, Gemini, etc.) you will be referred [Yale Executive Committee](#)

- Also, if you cheat on the homework you will do very poorly on the exams

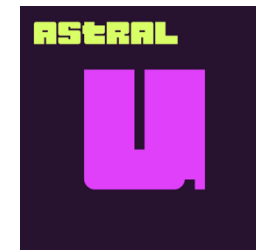
# Running Jupyter Notebooks

In order to do the homework, you will need to be able to run Jupyter Notebooks

The strongly preferred way to do this is to use the [YCRC Jupyter server](#)

Homework 0 allows you to test that you have a working Jupyter Notebook environment

- Homework 0 is not turned in, but please try it soon
- Ask questions on **Ed Discussions** or go to office hours to get help



# Let's test the YCRC Jupyter notebook server...

Before we get started, let's test the [YCRC Jupyter notebook server](#)

A link to the server is at the top of the class Canvas page

**If you can't log in, let us know on the background survey...**

Shivam will contact the YCRC and have you added to the server

# Class survey

In order to get to know you and to adjust the class to everyone's interests, please fill out the class survey on canvas

- Under the Quizzes link on the left

Any questions about the class logistics???

- Ask on Ed Discussions!



# What is Data Science?



What is Data Science?

**Data Science** is the discipline of turning raw data into **actionable knowledge**—using a blend of statistics, machine learning, computing, and domain understanding to answer questions, make predictions, and guide decisions.

**ChatGPT**

what is Data Science?



Data science is a multidisciplinary field that uses scientific methods, algorithms, and systems to extract knowledge and actionable insights from both structured and unstructured data. Think of it as a blend of **statistics, computer science, and business intuition**. [🔗](#)

As we move into 2026, data science has evolved from a "nice-to-have" technical advantage into a strategic necessity for nearly every industry, from healthcare diagnostics to urban planning in smart cities. [🔗](#)

**Gemini**

# Brief history of Data Science: data

## The first data we know of:

- The **Ishango bone** is a bone tool and possible mathematical device discovered at in the Democratic Republic of Congo
- Believed to about 20,000 years old

Cuneiform tablets from Uruk, a Mesopotamian settlement 5,000 years old contained transaction data on commodities



# Brief history of Data Science: demography and probability

John Graunt (1620-1674) develops statistical census methods that provided a framework for modern demography. He is credited with producing the first life table, giving probabilities of survival to each age



The mathematics of probability began to be developed in Europe starting in the 17<sup>th</sup> century

- Fermat and Pascal (1654), Bernoulli (1713), De Moivre (1718), Gauss and Laplace (1812)



# Brief history of Data Science: visualization and math

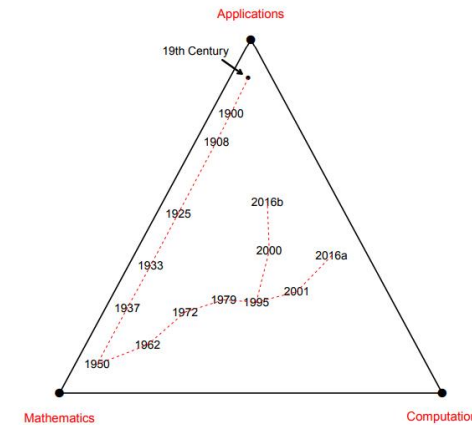
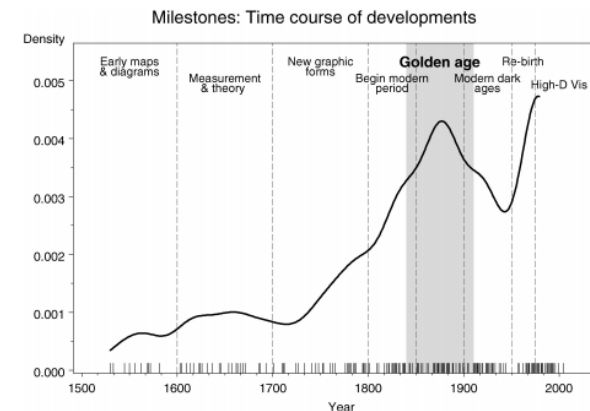
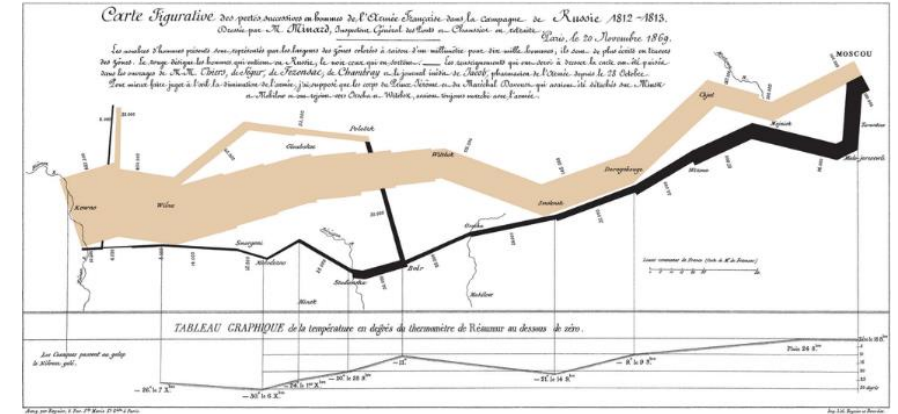
In the second half of the 19<sup>th</sup> century:

1. The field of Statistics uses probability models to analyze data
2. Elaborate visualizations of data were published

## Probability models dominate Statistics in the 20<sup>th</sup> century

Experimental data becomes dominant in the science and medicine in the second half of the 20<sup>th</sup> century

- E.g., Randomized Controlled Trials in Medicine



# Brief history of Data Science: the rise of computers

Early computational devices include:

- The abacus comes from Babylon in 2400 BCE
- Antikythera mechanism (~100 BCE) is an ancient Greek hand-powered device used to predict astronomical positions

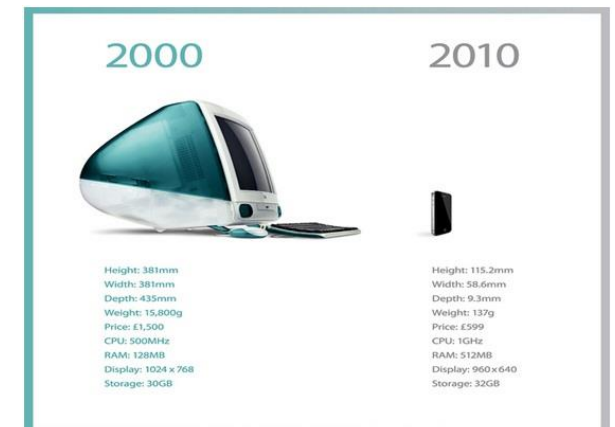


Herman Hollerith develops the Hollerith Tabulating Machine for the 1890 census (reduces 10 years of work to 3 months). Creates IBM



Computer technology develops rapidly in the second half of the 20<sup>th</sup> century

|        |                      |
|--------|----------------------|
| 1940's | Mainframe computers  |
| 1970's | Relational databases |
| 1980's | Personal computers   |
| 1989   | The world wide web   |
| 2007   | iPhone               |



# Brief history of Data Science: the rise of Data Science

The rise of powerful computers and plentiful data has given rise to new approaches to analyzing data

- John Tukey (1962) looks for a broadening of data analysis beyond mathematics
- Breiman (2001) describes a mathematical modeling culture and algorithmic culture
- The term "Data Science" starts being used in the 2000's to describe computational approaches to analyzing data
  - E.g., Cleveland 2001

## **THE FUTURE OF DATA ANALYSIS<sup>1</sup>**

BY JOHN W. TUKEY

*Princeton University and Bell Telephone Laboratories*

Statistical Science  
2001, Vol. 16, No. 3, 199–231

## **Statistical Modeling: The Two Cultures**

Leo Breiman

International Statistical Review (2001), 69, 1, 21–26, Printed in Mexico  
© International Statistical Institute

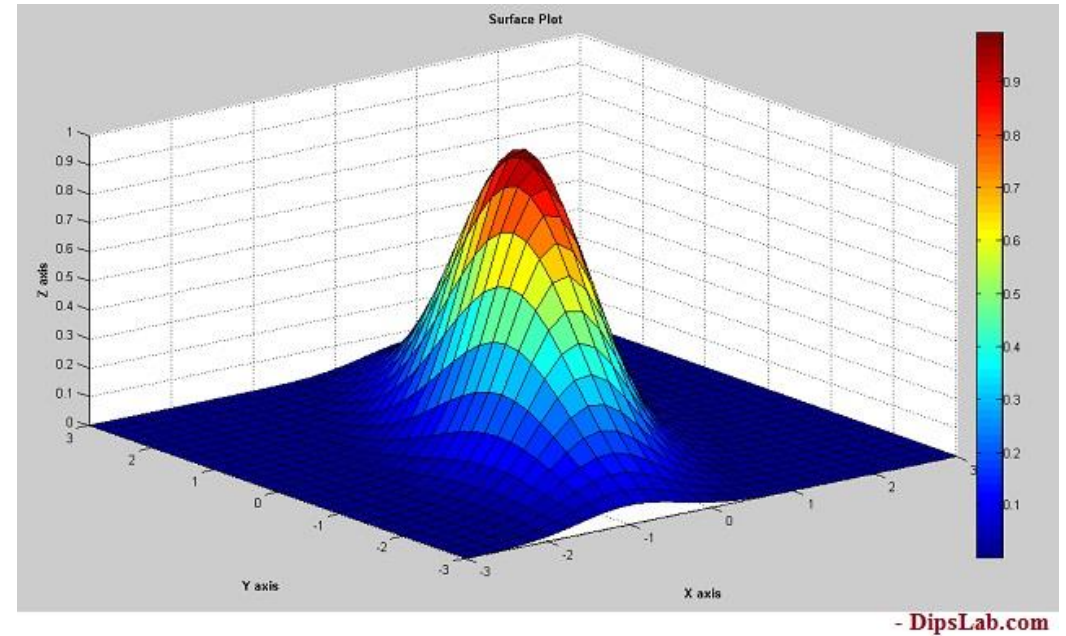
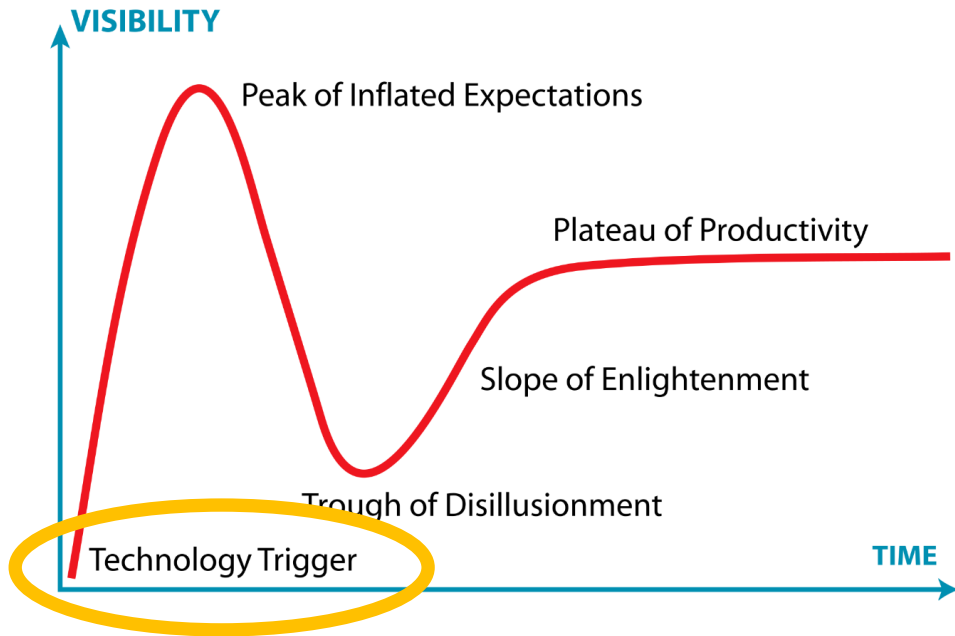
## **Data Science: an Action Plan for Expanding the Technical Areas of the Field of Statistics**

William S. Cleveland

Statistics Research, Bell Laboratories, 600 Mountain Avenue, Murray Hill NJ07974, USA  
E-mail: wsc@research.bell-labs.com



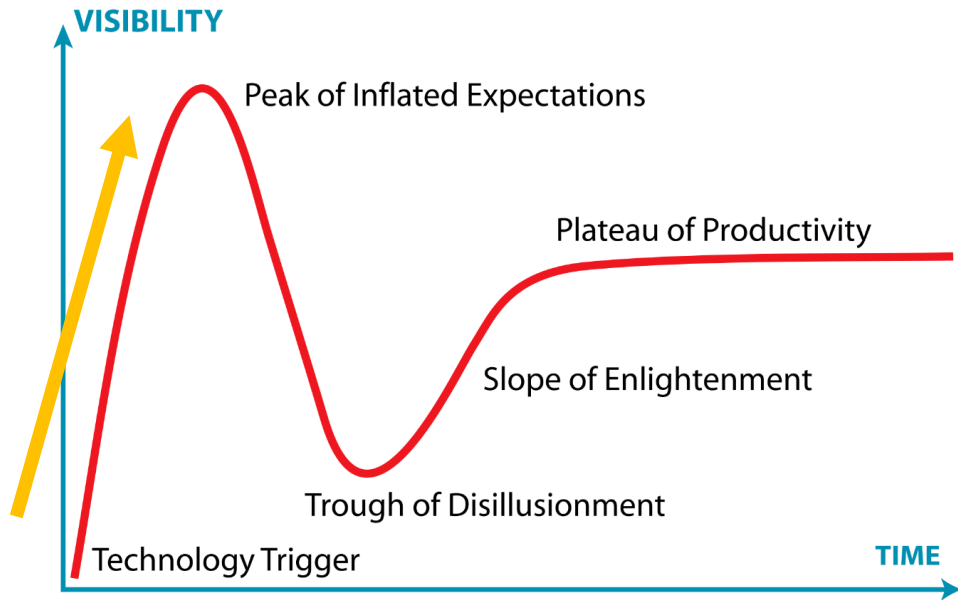
# Brief history of Data Science: Technology Trigger



Early 2000's rise of the internet, personal computers and data analysis programming languages changes how data can be analyzed

- MATLAB
- Python: matplotlib 2003, numpy 2005, scikit-learn 2007, pandas 2009

# Brief history of Data Science



Data Science rises with data science competitions, blog posts, and industry jobs

- Data Scientists viewed as “unicorns” because they had to know both statistics and how to program

2009



2010

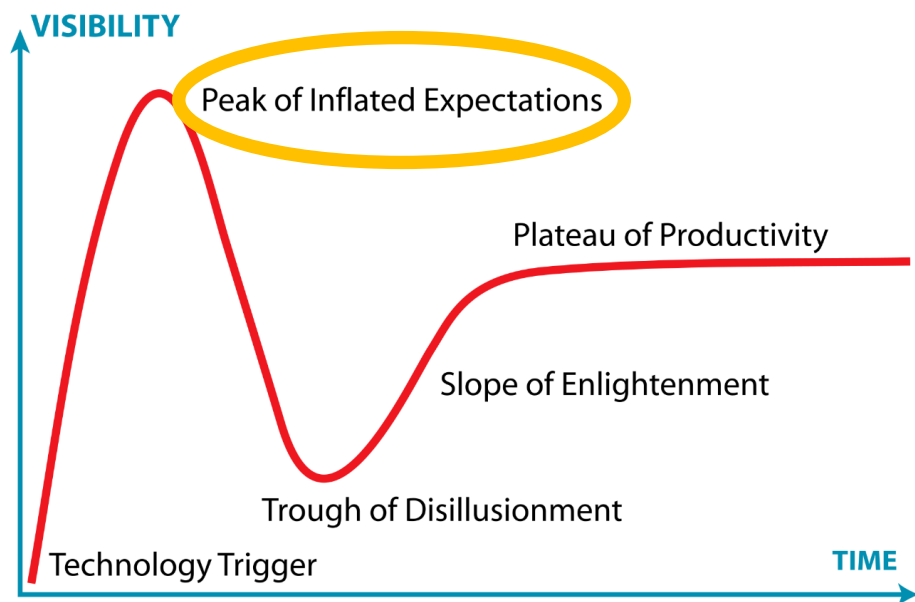
kaggle

Blog: 2009-2010

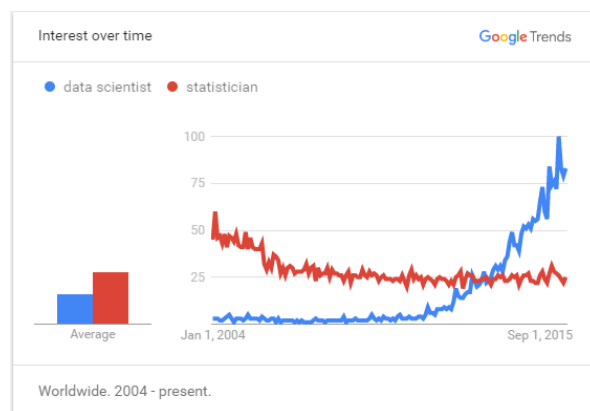
okcupid



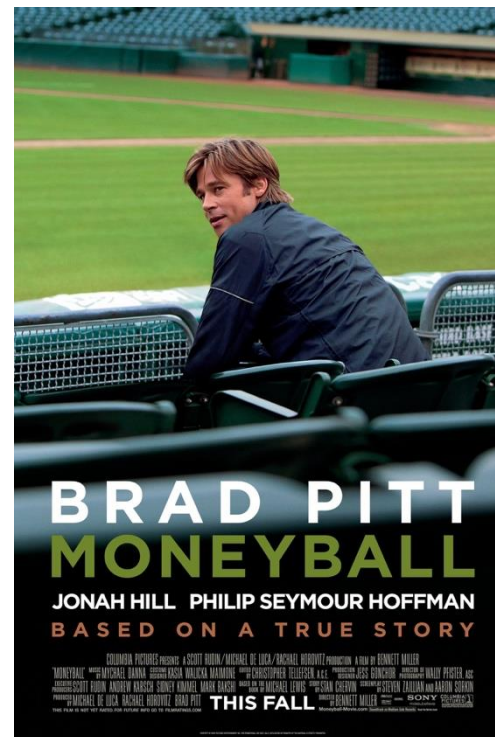
# Brief history of Data Science



Data Science hits “peak of inflated expectations” around 2012-14



2011



2012



Harvard Business Review

2012

Subscribe

Sign In

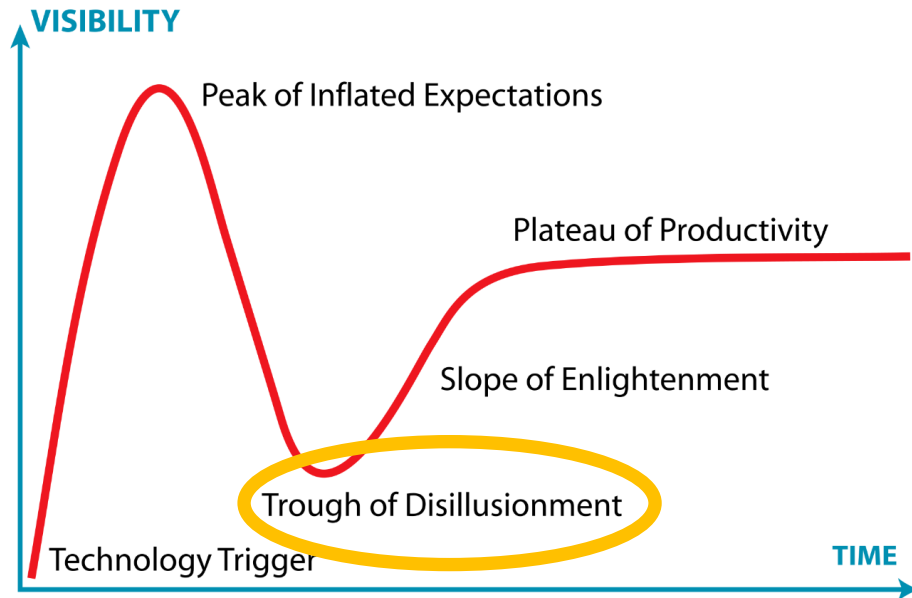
Latest Magazine Ascend Topics Podcasts Store The Big Idea Data & Visuals Case Selections

## Data Scientist: The Sexiest Job of the 21st Century

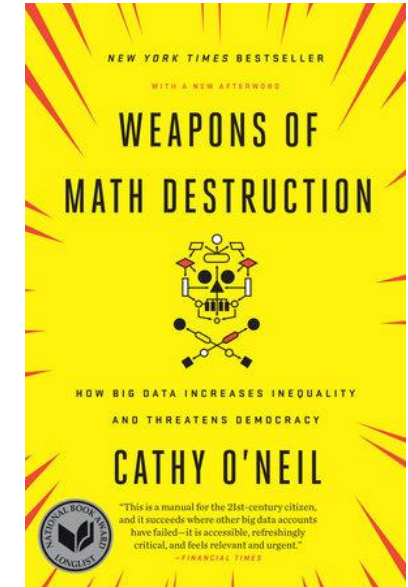
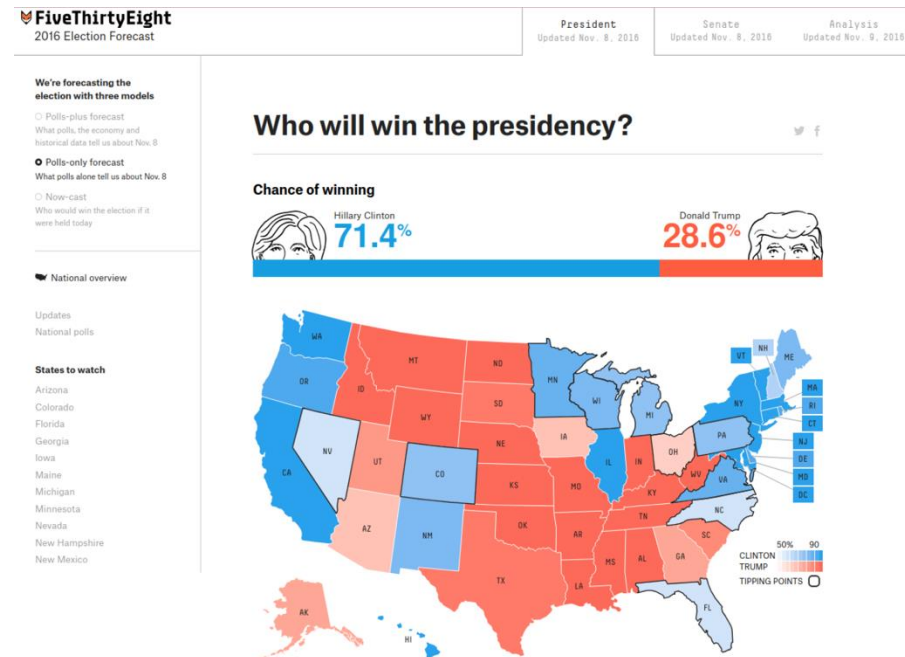
Meet the people who can coax treasure out of messy, unstructured data. by Thomas H. Davenport and DJ Patil

From the Magazine (October 2012)

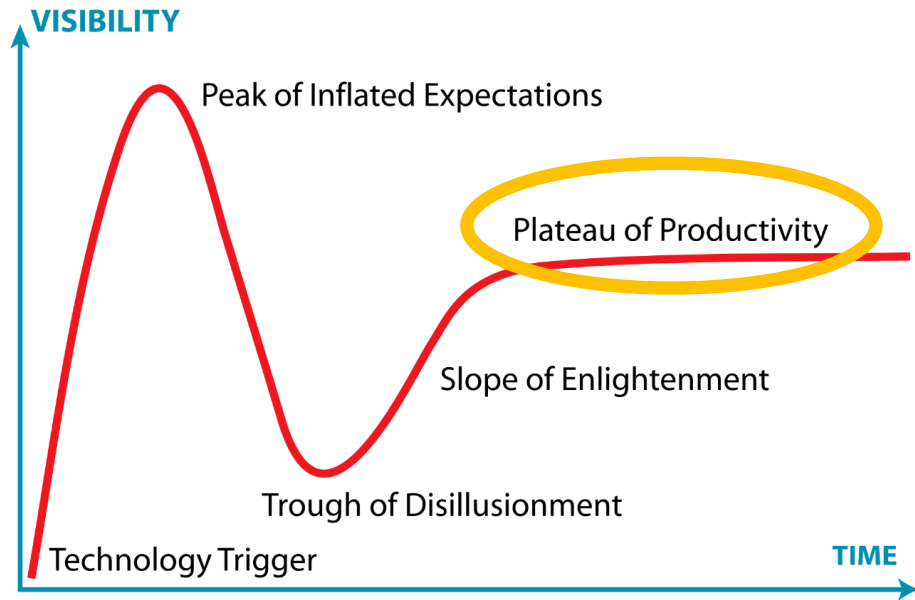
# Brief history of Data Science: Technology Trigger



Negative consequences of predictive models were highlighted circa 2016



# Brief history of Data Science: now



Yale  
S&DS

Data Scientists roles in many industries

- E.g., Data journalism ([NYTimes TheUpshot](#))

Many universities have Data Science programs

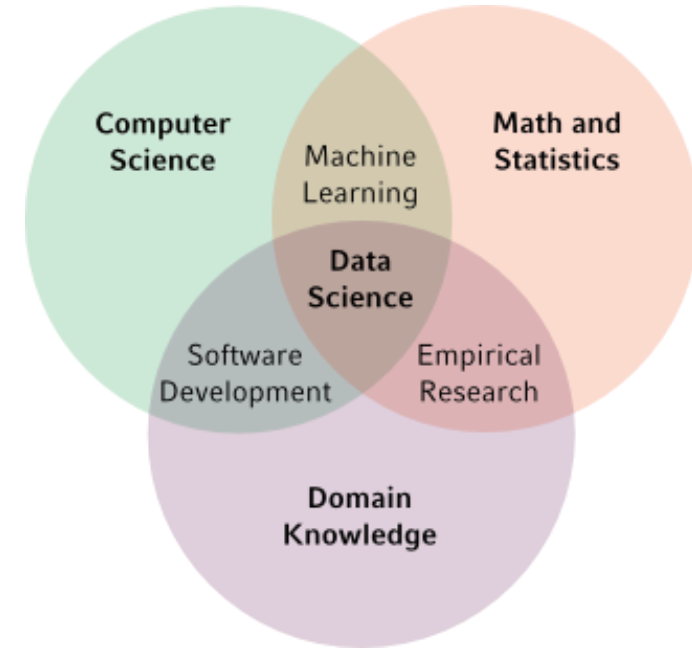
- In March 2017 Yale renames the Department of Statistics to be the Department of Statistics and Data Science



# What is Data Science?

Data Science is a broadening of data analyses:

- Beyond what traditional Statistical mathematical/inferential analyses
- To using more computation



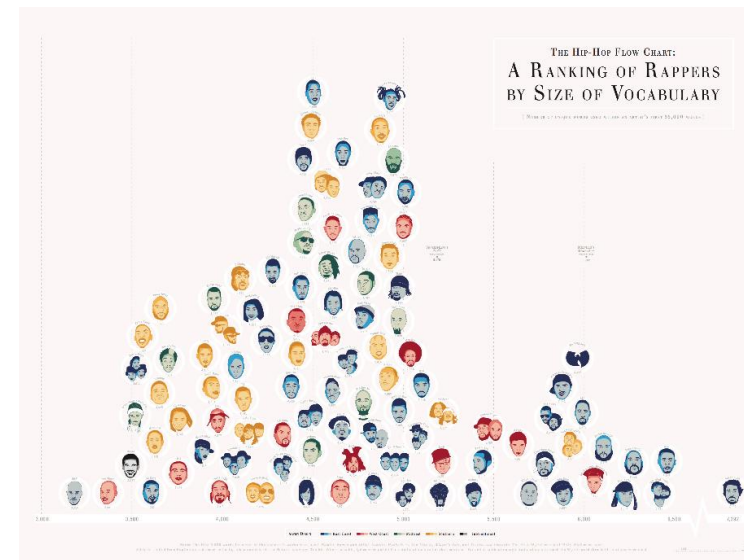
## Classical statistical analysis

Descriptives

| Bronchial reactivity |    |        |                |            |                                  |             |         |         |
|----------------------|----|--------|----------------|------------|----------------------------------|-------------|---------|---------|
|                      | N  | Mean   | Std. Deviation | Std. Error | 95% Confidence Interval for Mean |             | Minimum | Maximum |
|                      |    |        |                |            | Lower Bound                      | Upper Bound |         |         |
| Sulfur dioxide       | 5  | 18.840 | 9.6919         | 4.3344     | 6.806                            | 30.874      | 5.1     | 30.1    |
| Nitrous dioxide      | 6  | 6.617  | 3.9448         | 1.6105     | 2.477                            | 10.757      | 2.2     | 11.9    |
| Oxygen               | 4  | 4.975  | 3.4092         | 1.7046     | -.450                            | 10.400      | 2.1     | 9.3     |
| Total                | 15 | 10.253 | 8.6514         | 2.2338     | 5.462                            | 15.044      | 2.1     | 30.1    |

ANOVA

| Bronchial reactivity |                |    |             |       |      |
|----------------------|----------------|----|-------------|-------|------|
|                      | Sum of Squares | df | Mean Square | F     | Sig. |
| Between Groups       | 559.450        | 2  | 279.725     | 6.873 | .010 |
| Within Groups        | 488.408        | 12 | 40.701      |       |      |
| Total                | 1047.857       | 14 |             |       |      |



Examples:

- [NYC city bikes](#)
- [Wind map visualization](#)

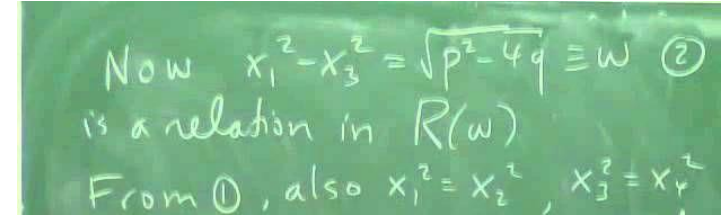


# New ways to choose the best methods

**Statistics** focuses on mathematical models (probability distributions) to analyze data

- Best methods are the ones that have mathematical guarantees (proofs)

**The proof is in the math**



Now  $x_1^2 - x_3^2 = \sqrt{p^2 - 4q} \equiv w$  ②  
is a relation in  $R(w)$   
From ①, also  $x_1^2 = x_2^2$ ,  $x_3^2 = x_4^2$

**Data Science** empirically evaluates data analysis methods

- Best methods are the one that gives the most insight in practice

**The proof is in the pudding**



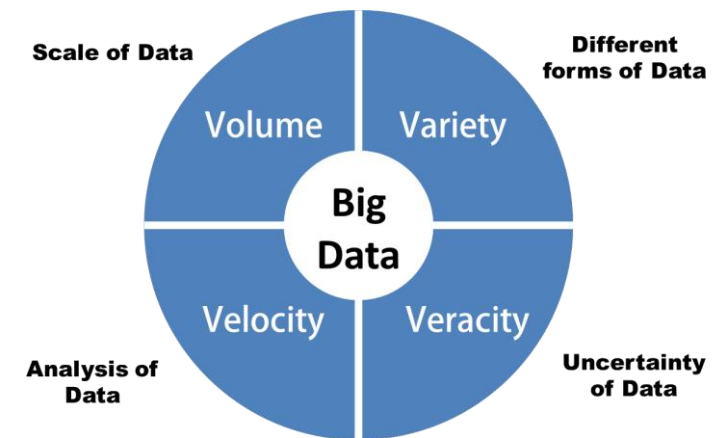
# Big Data

## New insights:

- Lots of new data from Internet, sensors etc., can be mined to transform our understanding in a range of fields
  - E.g., health, cosmology, social sciences, etc.

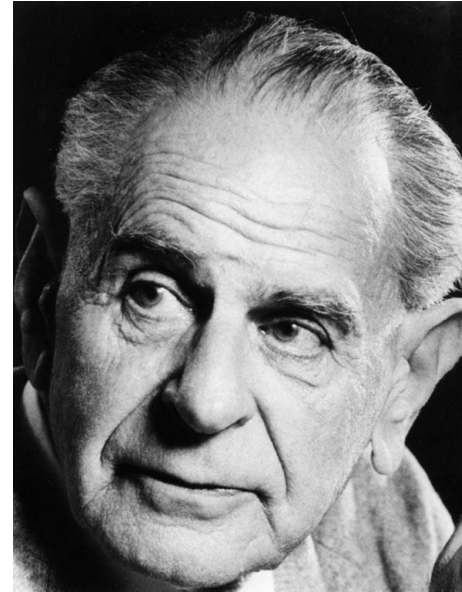
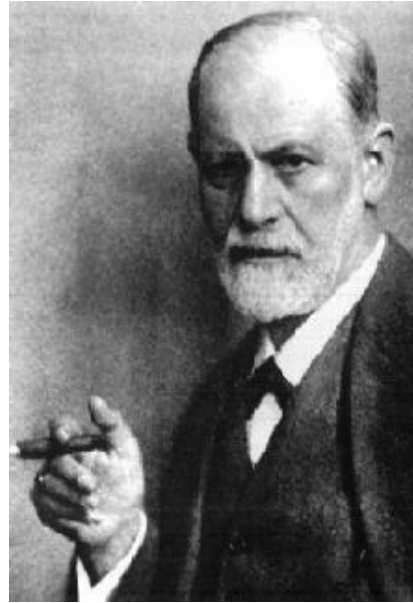
## New analysis and approaches:

- Hypothesis test pick up on very small (meaningless) effects with very large samples
- Data manipulation and programming are needed to extract insights
- Also, new standards for choosing the best data analysis methods



[Data Science vs. Statistician video](#)

# Short paper to read from the book Everybody Lies



Much of Freud's theory dealt with the subconscious

- E.g., Freudian slips

Karl Popper claimed that Freud's theories were unscientific because they couldn't be falsified

- i.e., can come up with any 'just so' story to explain a behavior

New data science analyses might make it possible to actually test Freud's theories

# Things to do for next class...

1. Complete class survey
2. Do short reading on Canvas from the book "Everybody lies"
3. Make sure you can log into the YCRC Jupyter Server